INCH-POUND
MIL-PRF-29504/12A
w/AMENDMENT 1
30 April 2024
SUPERSEDING
MIL-PRF-29504/12A
12 November 2002

PERFORMANCE SPECIFICATION SHEET

TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENT RESISTING, PIN TERMINUS, FRONT RELEASE, STAINLESS STEEL (FOR MIL-PRF-28876 CONNECTORS)

INACTIVE FOR NEW DESIGN AFTER 15 JANUARY 2002

FOR NEW DESIGN, USE MIL-PRF-29504/14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-29504.

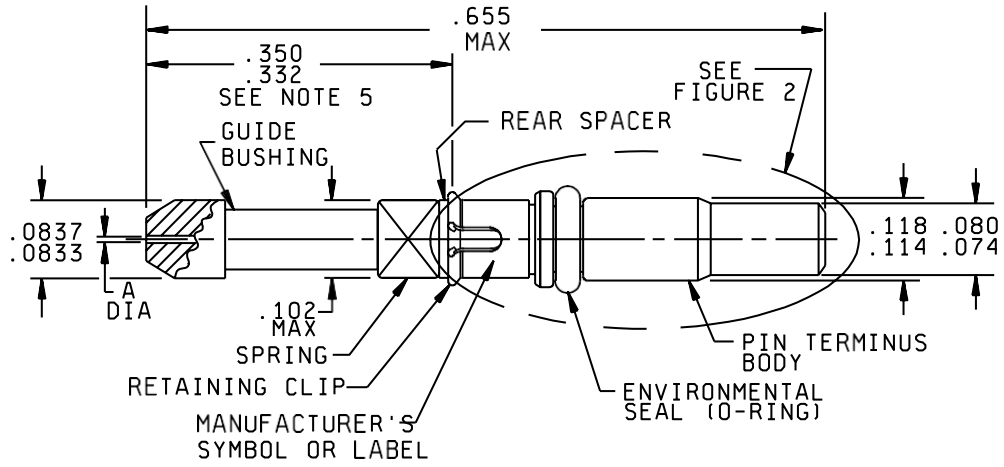


FIGURE 1. Pin terminus.

Distribution Statement A. Approved for public release: distribution is unlimited.

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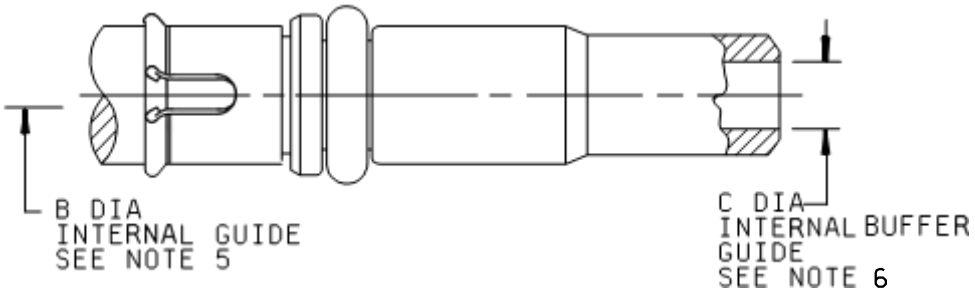
Inches	mm
.074	1.88
.080	2.03
.0833	2.116
.0837	2.126
.102	2.59
.114	2.90
.118	3.00
.332	8.43
.350	8.89
.665	16.89

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within 0.002 inch (0.05 mm).
4. Dimensions apply after plating.
5. Dimension .332 (8.43 mm) or .350 (8.89 mm) to be measured when installed in connector insert or equivalent gauge fixture. See figure 3.
6. The TICC bands shall not appear on terminus. Individual bag or container shall have either TICC bands or equivalent MIL-PRF-29504 Part or Identifying Number (PIN) marking on container or a tag inside container.

FIGURE 1. Pin terminus - Continued.

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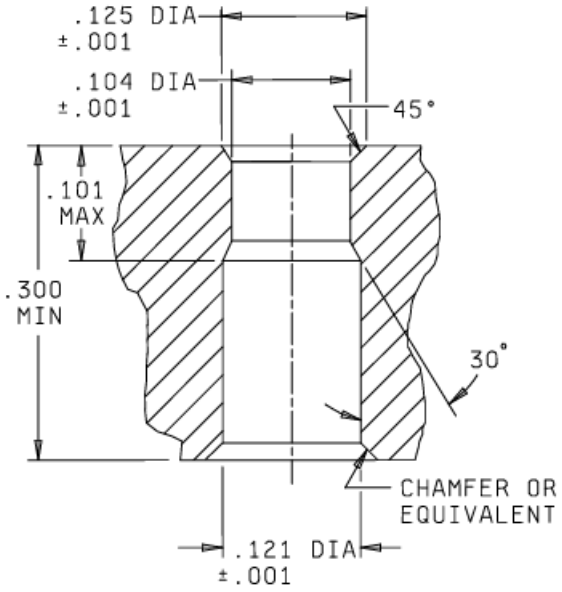
B diameter (internal guide)		C diameter (internal buffer guide)	
Inches	**mm**	**Inches**	**mm**
0.017 0.014	0.43 0.35	0.045 0.041	1.14 1.04

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .002 inch (0.05 mm).
4. Dimensions apply after plating.
5. The “B” diameter of the internal guide is used to center coated optical waveguide fiber.
6. The “C” diameter of the internal buffer guide is used to center the optical waveguide fiber buffer.

FIGURE 2. Terminus internal guide and buffer.

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Inches	mm
.001	0.03
.101	2.57
.104	2.64
.121	3.07
.125	3.18
.300	7.62

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .002 inch (0.05 mm).
4. Dimensions apply after plating.
5. Tolerance on all angles is ±1°, unless otherwise noted.

FIGURE 3. Insert equivalent fixture for terminus measurement.

REQUIREMENTS:

Temperature: -28 °C to +65 °C operating, -40 °C to +70 °C non-operating, -40 °C to +70 °C storage.

Design and construction:

Dimensions and configuration: See figures 1 and 2 and table I.

Weight: 1.0 gram maximum.

Adhesives: Use MIL-PRF-24792 or as approved by the qualifying activity.

Tools: See table II.

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Mating termini: MIL-PRF-29504/2 and MIL-PRF-29504/13.

Crimp sleeve: (for 2.4 mm maximum diameter cable) to be supplied with terminus when specified in the PIN.

Circular runout: Not greater than 2.5 micrometers.

Optical performance:

Insertion loss: The initial insertion loss of a mated pin and socket shall be not greater than 1.5 dB. The maximum insertion loss of a mated pin and socket at any time during testing shall be not greater than 2.0 dB.

Environmental/mechanical: Termini shall be tested to the following MIL-PRF-28876 environmental and mechanical requirements. Change in optical transmittance and optical discontinuity requirements shall be as specified in MIL-PRF-28876.

Impact

Vibration

Shock

Thermal shock

Temperature/humidity cycling

Temperature cycling

Temperature life

Flammability

Ozone exposure

To qualify or requalify termini to this specification sheet, all requirements of MIL-PRF-28876 shall be met using the qualifying terminus in a connector qualified to MIL-PRF-28876.

Qualification connector: The qualification connector for this terminus shall be a qualified MIL-PRF-28876 connector.

Test specimens: Test specimens shall be constructed using a 62.5/125 micron optical fiber within a single fiber cable with an outer diameter not greater than 2.4 mm.

Part or identifying number (PIN): See table I and 6.6 of MIL-PRF-29504.

M29504/12	**XXXX**	**X**
Basic specification and specification sheet	TICC Code (see table I)	Crimp sleeve (optional) C = Crimp sleeve provided Blank = Crimp sleeve not provided

Usage: Termini compliant with this specification sheet may be used in connectors other than MIL-PRF-28876 at the discretion of the acquiring activity.

Qualification by similarity: Multimode termini shall be granted qualification upon the successful qualification of single mode termini.

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TABLE I. TICC numbers to fiber diameter cross reference.

TICC number		A diameter (hole diameter) 1/	
Multimode	**Single-mode**	**Inches +0.0001 -0.0000**	**µm 2/ +2.5 -0.0**
4087	4097	0.0048	122.0
4088	4098	0.0049	124.5
4089	4099	0.0050	127.0
4090	4100	0.0051	129.5
4091	4101	0.0054	137.0
4092	4102	0.0055	139.5
4093	4103	0.0056	142.0
4094	4104	0.0057	145.0
4095	4105	0.0091	231.5
4096	4106	0.0092	234.0

NOTES:
1/ For maximum performance, termini should be selected for the best (tightest) fit to the fiber. Fitting the next size larger terminus to a fiber may affect connector performance.
2/ Rounded to the nearest 0.5 µm.

TABLE II. Tools.

Tool	Part number
Insertion tool	NAVSEA DWG 6872813-2 (NSN 5120-01-144-5338)
Removal tool	NAVSEA DWG 6872813-6 (NSN 5120-01-419-2942)
Polishing tool	Packard Hughes PN 4569100H or equivalent

Patent notice: The Government does not have royalty-free license under the following patents for the benefit of manufacturers of the item, either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
US 4707068	11/17/2004

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
Navy – SH
(Project 6060-2023-018)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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